Sector Trading and Bubble Riding Behavior: Construction and Evaluation of Trading Strategy Based on Bubble Test

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Abstract: In this paper, we investigate the profitability of "riding the bubble" trading strategies in the Chinese stock market. Using sector-level data from the CSI 300, we identify bubble episodes using the bubble test proposed by Phillips et al. (2015). We then buy the sector-level indices whenever bubbles are identified, and buy the market index otherwise. Empirical studies show that most riding strategies outperform the market in terms of annualized return and Sharpe ratio. In particular, the optimal "riding bubble" trading strategy can generate an average annualized return of 13.41% over the period 2006-2020, while the buy-and-hold strategy yields 11.87%.

Keywords: Bubble test; Trading strategy; GSADF test; PSY method; Sector-level market

JEL Classification: C12, C22, G01, G11

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1 Introduction

Stock market bubbles have provided a significant financial risk and resulted in heavy losses to global investors in history, making the reaction of investors facing bubbles a focal point of attention for academics. Fama (2014) states that the studies on bubbles (Shiller, 2000; Kindleberger et al., 2005) rely on beliefs that stock price declines are predictable and argues that stock prices are unpredictable and that rational investors correct mispricing by arbitrage. However, Greenwood et al. (2019) report that sharp

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price increases predict heightened probabilities of crashes, and investors can identify bubbles by some stock characteristics containing information relevant to forecasting expected returns. Abreu and Brunnermeier (2003) even suggest that the optimal choice for investors discovering a bubble is to ride it. This paper will address whether we can predict price dynamics to bet against bubbles. We construct effective bubble-riding strategies based on advanced bubble detection techniques in the Chinese stock market. If the trading strategies consistently yield abnormal returns, it's feasible for investors to identify bubbles and benefit from riding bubbles.

According to Temin and Voth (2004), Hoare's Bank successfully rode the South Sea Bubble by trading the overvalued stocks of South Sea Company. Brown et al. (2000), Brunnermeier and Nagel (2004), and Fung and Hsieh (2000) report that hedge funds profited from bubbles by holding portfolios heavily tilted toward highly-priced technology stocks during the technology bubble period. Brooks and Katsaris (2005) use a three-regime model to study the bubble dynamics of the S&P 500 and form a bubble trading rule based on this model. Guenster and Kole (2013) examined investors' decisions in the face of bubbles, proving that even individual investors may profit from investing in bubbles.

Our study begins with the bubble identification of the GSADF test on the 2006-2020 China Securities Index 300 (CSI300) market, including the most representative Chinese sample stocks. The GSADF test has been proven more effective than the SADF test of Phillips et al. (2011) and the methods of Hall et al. (1999) and Guenster et al. (2009). Then we construct three bubble-riding strategies based on the GSADF test, Strategies

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1, 2, and 3, which buy bubble portfolios, sell them and invest in the market portfolio when bubbles collapse according to their rules. To avoid daily price jumps, manage portfolios in time, and reduce idiosyncratic noise, we will conduct the above strategies in the sector markets at the weekly frequency and then explore their profitability against the benchmark (Strategy 4). The results show that our riding strategies perform better than the benchmark, indicating the possibility and profitability of riding bubbles in the Chinese market.

The paper contributes to the literature by examining the profitability of the "riding bubble" strategies in the Chinese sector-level stock market. These trading strategies try to construct a portfolio of stocks in the bubble periods and sell the overpriced stocks as soon as the bubbles explode. The profitable strategies show that Fama (2014)'s view doesn't hold in the Chinese market, and bubble riding is the rational behavior for Chinese investors facing bubbles. Additionally, the study unit of the paper is a sector rather than a firm. Compared with similar literature, this paper analyzes the industry characteristics of Chinese bubbles.

We organize the rest of the paper as follows. The next section introduces the methodologies. Section 3 describes our dataset and reports the sector-level bubble dynamics. Section 4 assesses the performance of all strategies. Section 5 provides the robustness checks. Section 6 concludes.

2 Methodology

2.1 The GSADF procedure for explosiveness

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The recursive flexible window procedure for identifying and dating bubbles called the GSADF test, is based on the following regression model.

$$\Delta y_t = \alpha_{r_1,r_2} + \beta_{r_1,r_2} y_{t-1} + \sum_{i=1}^{K} \delta^i_{r_1,r_2} \Delta y_{t-i} + \varepsilon_t \quad (1)$$

$$H_0{:}\beta_{r_1,r_2} = 0$$

$$H_A{:}\beta_{r_1,r_2} > 0$$

where y_t is the industry price index of interest, K is the lag order (selected by BIC with a maximum lag order of six), and ε_t is the error term.

GSADF test implements ADF regression (1) in a rolling window style, which requires running subsample regressions. Here the fractional starting and ending points of a subsample regression are r_1 and r_2, and the size of the total sample equals *T*, where the corresponding ADF statistic is $ADF_{r_1}^{r_2}$. Accordingly, the window size $r_w = r_2 - r_1$ and the number of observations equals $\lfloor r_w T \rfloor$.

We construct a sequence of subsamples by allowing r_2 to vary from r_0 to 1 and r_1 to vary from 0 to $r_2 - r_0$ (r_0 is the minimum window size equaling $0.01 + 1.8/\sqrt{T}$), then calculates the ADF statistic repeatedly on these subsamples, and defines the GSADF statistic as the largest one among the sequence of ADF statistics.

$$GSADF(r_0) = \sup_{r_2 \in [r_0,1]} \sup_{r_1 \in [0, r_2 - r_0]} \left\{ ADF_{r_1}^{r_2} \right\} \quad (2)$$

The backward sup ADF test (BSADF test) is used to time the origination and the termination of bubbles, which implements the ADF test repeatedly on a sequence of backward expanding samples. Assuming r is the observation of interest, we will fix the ending point of each sample r_2 on r and vary the start point r_1 from 0 to $r - r_0$. Then $MADF_r$ is defined as the supremum of the sequence $\left\{ ADF_{r_1}^{r_2} \right\}_{r_1 \in [0, r_2 - r_0]}^{r_2 = r}$, which

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identifies explosiveness at *r*. Due to the heteroskedasticity of weekly data, we will produce consistent critical values using a wild bootstrapping procedure of Harvey et al. (2016) and Phillips and Shi (2020).

$$MADF_r = \max\left\{ADF_{r_1}^{r_2}: r = r_2, r_1 \in [0, r - r_0]\right\} \quad (3)$$

2.2 Construction of trading strategies

We assume that an investor trades at a weekly frequency, observing the bubble signal at the end of Tuesdays and executing his trading decision on Wednesdays.

Referring to Milunovich et al. (2019), we propose three bubble-riding strategies, Strategy 1, Strategy 2, and Strategy 3, against our benchmark (Strategy 4, i.e., buy-and-hold strategy). The indicators of Strategies 1, 2, and 3 are PSYI (PSY Indicator), MPSYI (Modified PSY Indicator), and DPSYI (Dual PSY Indicator).

$$PSYI_t = 1(MADF_t > scv_t) \quad (4)$$

$$MPSYI_t = 1(MADF_t > scv_t \& R_t > 0) \quad (5)$$

$$DPSYI_t = 1(MADF_t > scv_t \& MADF_{t-1} > scv_{t-1}) \quad (6)$$

In the equations of the weekly indicators, *t* denotes time measured in weeks, $MADF_t$ is the statistic of the GSADF test for identifying bubbles in industry indexes, scv_t is the critical value of $MADF_t$, and R_t is the most recent return.

2.2.1 Rules of bubble-riding strategies

Strategy 1 buys any sector for which the GSADF test detects a bubble episode (the PSYI equals 1) and closes the positions when the bubble collapses (the PSYI switches from 1 to 0).

Strategy 2 uses the MPSYI instead of the PSYI. Specifically, Strategy 2 invests in the sectors which experience bubbles and generate positive recent returns (the MPSYI

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equals 1) and sells those sectors' stocks if the bubbles burst or the recent returns turn negative (the MPSYI equals 0).

Strategy 3 buys those sectors where the bubbles continuously exist at both the previous and current periods (the DPSYI equals 1) and exits from the collapsing bubbles (the DPSYI turns 0).

If there are multiple sectors satisfying their conditions at the same time, these strategies will construct an equally-weighted portfolio. In the event of no bubbles, they hold the market index.

2.2.2 Analysis of trading strategies

All bubble riding strategies try allocating the funds to the bubble sectors. Strategy 1 relies only on the PSY method (i.e., the GSADF test) to time bubble sectors. However, the GSADF test can't always keep bubble detection accurate. The false identification influences Strategy 1's performance adversely. To reduce the impact of the test bias, Strategy 2 exploits the MPSYI that introduces the recent return information. Strategy 3 raises the investment threshold to avoid entering a sector too early by observing additionally the bubble signal at the previous period.

3 Bubble dynamics in China's industry market

3.1 Data description

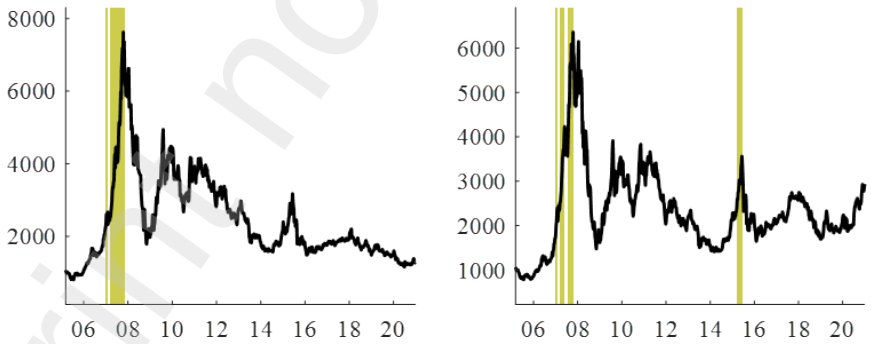
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The sector indexes of CSI 300 in the paper include energy, materials, industrials, consumer discretionary, consumer staples, health care, financials, information technology, communication services, and utilities. The dataset contains the following weekly data series (the sample period from January 4th, 2006, to December 30th, 2020): close prices of the sector indices on Tuesdays and Wednesdays; close prices of the CSI300 market index on Wednesdays; weekly CPI by the Denton (1971) method. In addition, the daily price series of the same indexes are applied in the robustness analysis to test the superiority of the weekly data frequency.

3.2 Empirical bubble detection

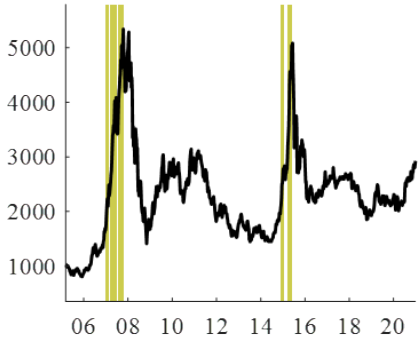
This paper applies the GSADF test at the 95% confidence level to detect explosive bubble dynamics. In Figure 1, the curves represent the industry prices, the horizontal axis marks the year, and the shaded areas correspond to all bubble dynamics we detect.



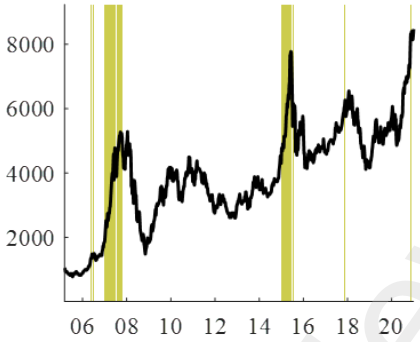
(a) Energy (b) Materials

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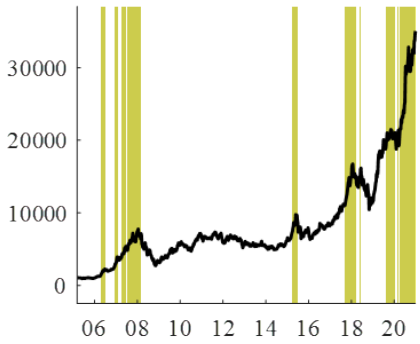
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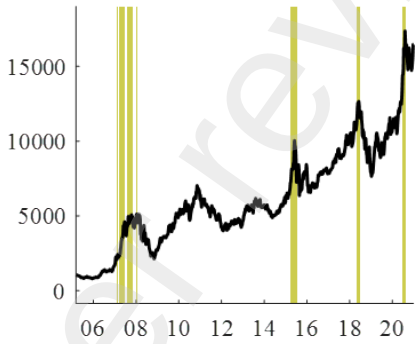
(c) Industrials



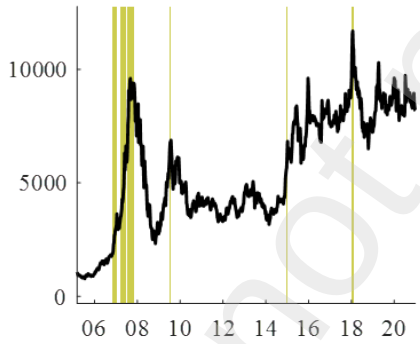
(d) Consumer discretionary



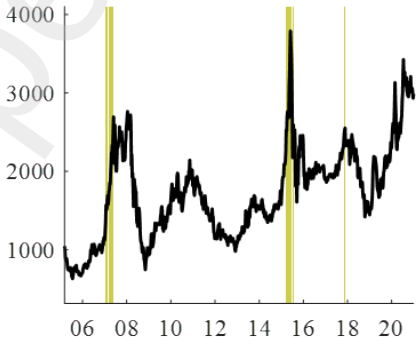
(e) Consumer staples



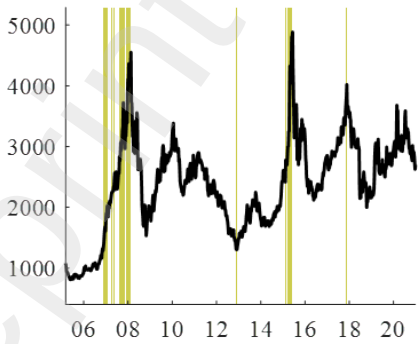
(f) Health care



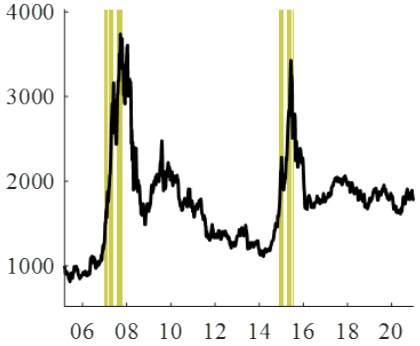
(g) Financials



(h) Information technology



(i) Telecom services



(j) Utilities

Figure 1 Bubble identification for each sector

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First, all sectors experienced bubble episodes during the sample period, but there are also some differences in the characteristics of these bubbles (frequency, duration, and degree of expansion). Second, the order of the origination dates of bubbles in these sectors suggests the contagion of bubble dynamics between them. The bubbles in industries like financials formed early and spread to other sector-level markets, providing a significant spillover risk. Third, there exists a covariation of the bubbles in ten sectors. The co-movement of bubble dynamics depicts the general trend of the aggregate market.

4 Performance evaluation of trading strategies

Table 1 displays the investment performance of four strategies on the following measures: final wealth accumulated over 2006-2020 with an initial wealth of $1, after transaction cost returns, and Sharpe ratios. According to Brooks and Katsaris (2005) and the reality of the Chinese market, a round trip of buying and selling incurs a trading cost of 0.5% of the total real return.

Table 1 Performance evaluation of trading strategies

Performance Measure	Strategy 1 (PSYI)	Strategy 2 (MPSYI)	Strategy 3 (DPSYI)	Strategy 4 (BH)
Final wealth($)	6.39	5.02	6.64	5.40
Annual return(%)	13.13	11.32	13.41	11.87
Sharpe ratio(%)	6.55	5.80	6.66	6.13
Trading periods	122	168	87	2

Strategy 3 performs best according to all investment criteria. Strategy 1 produces the second-highest wealth. The buy-and-hold benchmark is ranked third, which follows Strategies 1 and 3. Strategy 2 is ranked last and falls behind Strategies 1, 3, and 4. The performance of Strategies 1 and 3 supports the idea of profiting from inflating Chinese

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sector bubbles. Our best strategy, Strategy 3, invests in the sectors which have experienced a bubble for at least two periods (weeks), which prevents investors from initiating their bubble trades prematurely.

In conclusion, Strategy 3 can generate the highest return, beat the comparable benchmark, and benefit more effectively from bubbles in weekly trading.

5 Robustness analysis

5.1 Subsample analysis

By a rolling window subsample analysis, this section explores the consistency of these strategies' returns over different holding periods. We assume an investor enters the market at any time and invests for *k* periods (weeks). The investment horizon is represented by a window of width *k* periods (weeks), which is rolled by one week at a time.

Focusing on Strategy 3, Strategy 1, and the benchmark, we assess their relative performances over a range of short subsamples by varying the window width *k* between 1 and 6 years ($k = 1 \times 52, 2 \times 52, \cdots, 6 \times 52$ weeks). Now we will compare the annual returns of any two strategies over every subsample. For instance, when comparing Strategy 1 with Strategy 3, we calculate the fraction of *k*-weeks subsamples over which Strategy 1 outperforms, underperforms, and performs equally to Strategy 3.

5.1.1 Strategy 1 (PSYI) versus Strategy 3 (DPSYI)

The outperformance percentage of Strategy 3 is higher than that of Strategy 1 across any rolling window. Because Strategy 3 can better reduce the impact of false detection

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on strategies' returns over a long sub-sample period, it will generate an advantage over Strategy 1 as the window size increases.

Table 2 The relative rolling window performance of strategy 1(PSYI) and strategy 3(DPSYI)

Case	DPSYI outperforms (PSYI underperforms)		PSYI outperforms (DPSYI underperforms)		Equal Performance
Window Size	Percentage	Av.return diff.	Percentage	Av.return diff.	Percentage
One year	38.03%	4.79%	26.68%	5.07%	35.29%
Two years	63.62%	1.79%	24.30%	2.50%	12.08%
Three years	71.13%	1.28%	25.68%	2.17%	3.19%
Four years	73.04%	1.07%	26.96%	1.44%	0%
Five years	76.10%	0.90%	23.90%	1.09%	0%
Six years	77.71%	0.81%	22.29%	0.58%	0%

5.1.2 Strategy 3 (DPSYI) versus Strategy 4 (BH)

Table 3 The relative rolling window performance of strategy 3(DPSYI) and strategy 4(BH)

Case	DPSYI outperforms (BH underperforms)		BH outperforms (DPSYI underperforms)		Equal Performance
Window Size	Percentage	Av.return diff.	Percentage	Av.return diff.	Percentage
One year	36.39%	9.12%	28.32%	7.97%	35.29%
Two years	47.28%	5.42%	41.09%	3.35%	11.63%
Three years	53.91%	2.90%	43.22%	2.10%	2.87%
Four years	57.57%	2.03%	42.43%	1.77%	0%
Five years	61.76%	1.73%	38.24%	1.73%	0%
Six years	64.12%	1.54%	35.88%	1.61%	0%

It is necessary to compare Strategy 3 with the benchmark. Across all window lengths, Strategy 3 is more likely to outperform rather than underperform Strategy 4. As the window size increases, the subsample contains more bubbles identified by the GSADF test, and Strategy 3 can perform better by riding explosive bubbles.

5.2 Switching data frequency

This paper will employ sensitive daily data instead to explore the feasibility of daily data, assuming investors make trading decisions every day. Table 4 demonstrates that

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the performance of the strategies is poor in daily sector trades. The volatile daily data can reduce the accuracy of bubble detection and increase trading costs.

Table 4 Performance evaluation of trading strategies with daily data

Performance	With transaction costs			Without transaction costs			
Evaluation	PSYI	MPSYI	DPSYI	PSYI	MPSYI	DPSYI	BH
Final wealth ($)	0.70	0.02	1.37	7.81	9.68	7.90	5.40
Annual return (%)	-3.30%	-29.76%	3.01%	21.15%	23.60%	21.28%	11.87
Sharpe ratio (%)	-0.08%	-5.12%	0.9%	3.41%	3.74%	3.42%	6.13
Trading periods	481	1207	348	481	1207	348	2

5.3 Trading at different confidence levels

Now we will explore the influence of different confidence levels on Strategy 3's performance by using 90% and 99% alternative confidence levels. As the confidence level decreases from 99% to 90%, the returns of Strategy 3 will see an increase, followed by an evident decrease. A high confidence level increases the accuracy of the bubble test and results in a high identification threshold. A low confidence level detects more bubbles but leads to biases and high-frequency costs.

Table 5 Performance evaluation of Strategy 3 for alternative confidence levels

Performance Measure	DPSYI-90%	DPSYI-95%	DPSYI-99%
Final wealth($)	5.41	6.64	6.47
Annual return(%)	11.88	13.41	13.22
Sharpe ratio(%)	6.01	6.66	6.64

6 Conclusion

By employing the GSADF bubble test on ten CSI300 sector indexes, the paper identifies the existence of bubbles across ten sectors, demonstrates their differences in frequency, duration, and expansion degree, and indicates the contagion of bubble dynamics in these sectors.

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Based on the GSADF test, this paper constructs three bubble-riding strategies, Strategies 1, 2, and 3, and chooses the buy-and-hold strategy as the benchmark (Strategy 4). Strategy 3 performs best with transaction costs, followed by Strategy 1, Strategy 4, and Strategy 2. Then our subsample analysis verifies the consistency in Strategy 3's performance by assessing its return over different periods. To confirm the robustness of our findings, we also take other measures like using daily data and changing the confidence of the bubble test.

The profitability of Strategy 3 effectively demonstrates the rationality of riding bubbles in the Chinese sector markets. The bubble-riding strategies based on the GSADF test can beat the benchmark and yield high abnormal returns, showing that rational investors may drive the expansion of bubbles.

Declaration of competing interest

The authors declare that there is no potential conflict of interest.

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